Timers in LPC2148



Features of Timer in LPC2148

- TimerCounter0 and TimerCounter1 are functionally identical.
- Each 32-bit Timer/Counter has a programmable 32-bit Prescaler.
- Counter or Timer operation.
- Up to four 32-bit capture channels per timer,
  - that can take a snapshot of the timer value when an input signal transitions.
  - A capture event may also optionally generate an interrupt.
- Four 32-bit match registers that allow:
  - Continuous operation with optional interrupt generation on match.
  - Stop timer on match with optional interrupt generation.
  - Reset timer on match with optional interrupt generation.
- Up to four external outputs corresponding to match registers, with the following capabilities:
  - Set low on match.
  - Set high on match.
  - Toggle on match.
  - Do nothing on match.

CTCR : Count Control register

- Used to select Timer/Counter Mode.
- For Timing purpose, we use this in Timer Mode.
- When the value of the CTCR is set to 0×0 Timer Mode is selected.

Bit	Symbol	Value	Description	Reset value
1:0	Counter/ Timer Mode		This field selects which rising PCLK edges can increment Timer's Prescale Counter (PC), or clear PC and increment Timer Counter (TC).	00
		00	Timer Mode: every rising PCLK edge	
		01	Counter Mode: TC is incremented on rising edges on the CAP input selected by bits 3:2.	
		10	Counter Mode: TC is incremented on falling edges on the CAP input selected by bits 3:2.	
		11	Counter Mode: TC is incremented on both edges on the CAP input selected by bits 3:2.	

Prescaler

- PR : Prescale Register (32 bit) –Stores the maximum value of Prescale counter after which it is reset.
- PC : Prescale Counter (32 bit) –This register increments on every PCLK(Peripheral clock). This register controls the resolution of the timer.
- When PC reaches the value in PR , PC is reset back to 0 and Timer Counter is incremented by 1.
- Hence if PR=0 then Timer Counter Increments on every 1 PCLK.
- If PR=9 then Timer Counter Increments on every 10th cycle of PCLK.
- Hence by selecting an appropriate Prescale value we can control the resolution of the timer.

F out = Fin / (PR + 1)

TC : Timer Counter Register (32 bit)

- This is the main counting register.
- Timer Counter increments when PC reaches its maximum value as specified by PR.
- If timer is not reset explicitly(directly) or by using an interrupt then it will act as a free running counter which resets back to zero when it reaches its maximum value which is 0xFFFFFFFF.

Total Counts = **429,49,67,296.**

TCR : Timer Control Register

- This register is used to enable , disable and reset TC.
- When bit0 is 1 timer is enabled and when 0 it is disabled.
- When bit1 is set to 1 TC and PC are set to zero together in sync on the next positive edge of PCLK.
- Rest of the bits of TCR are reserved.

Bit	Symbol	Description	Reset value
0	Counter Enable	When one, the Timer Counter and Prescale Counter are enabled for counting. When zero, the counters are disabled.	0
1	Counter Reset	When one, the Timer Counter and the Prescale Counter are synchronously reset on the next positive edge of PCLK. The counters remain reset until TCR[1] is returned to zero.	0

MCR : Match Control register

- This register is used to control which all operations can be done when the value in MR matches the value in TC.

- Bits 0,1,2 are for MR0 , Bits 3,4,5 for MR1 and so on. Here is a quick table which shows the usage:

➢ Bit 0 : Interrupt on MR0 i.e trigger an interrupt when MR0 matches TC. Interrupts are enabled when set to 1 and disabled when set to 0.

➢ Bit 1 : Reset on MR0. When set to 1 , TC will be reset when it matched MR0. Disabled when set to 0.

➢ Bit 2 : Stop on MR0. When set to 1 , TC & PC will stop when MR0 matches TC.

T0MCR

Bit	Symbol	Value	Description	Reset value
0	MR0I	1	Interrupt on MR0: an interrupt is generated when MR0 matches the value in the TC.	0
		0	This interrupt is disabled	
1	MR0R	1	Reset on MR0: the TC will be reset if MR0 matches it.	0
		0	Feature disabled.	
2	MR0S	1	Stop on MR0: the TC and PC will be stopped and TCR[0] will be set to 0 if MR0 matches the TC.	0
		0	Feature disabled.	
3	MR1I	1	Interrupt on MR1: an interrupt is generated when MR1 matches the value in the TC.	0
		0	This interrupt is disabled	
4	MR1R	1	Reset on MR1: the TC will be reset if MR1 matches it.	0
		0	Feature disabled.	
5	MR1S	1	Stop on MR1: the TC and PC will be stopped and TCR[0] will be set to 0 if MR1 matches the TC.	0
		0	Feature disabled.	
6	MR2I	1	Interrupt on MR2: an interrupt is generated when MR2 matches the value in the TC.	0
		0	This interrupt is disabled	
7	MR2R	1	Reset on MR2: the TC will be reset if MR2 matches it.	0
		0	Feature disabled.	
8	MR2S	1	Stop on MR2: the TC and PC will be stopped and TCR[0] will be set to 0 if MR2 matches the TC.	0
		0	Feature disabled.	
9	MR3I	1	Interrupt on MR3: an interrupt is generated when MR3 matches the value in the TC.	0
		0	This interrupt is disabled	
10	MR3R	1	Reset on MR3: the TC will be reset if MR3 matches it.	0
		0	Feature disabled.	
11	MR3S	1	Stop on MR3: the TC and PC will be stopped and TCR[0] will be set to 0 if MR3 matches the TC.	0

IR : Interrupt Register

- It contains the interrupt flags for 4 match and 4 capture interrupts.
- Bit0 to bit3 are for MR0 to MR3 interrupts respectively.
- And similarly, the next 4 for CR0-3 interrupts.
- when an interrupt is raised the corresponding bit in IR will be set to 1 and 0 otherwise.
- Writing a 1 to the corresponding bit location will reset the interrupt – which is used to acknowledge the completion of the corresponding ISR execution.

T0IR : Interrupt Register

Bit	Symbol	Description	Reset value
0	MR0 Interrupt	Interrupt flag for match channel 0.	0
1	MR1 Interrupt	Interrupt flag for match channel 1.	0
2	MR2 Interrupt	Interrupt flag for match channel 2.	0
3	MR3 Interrupt	Interrupt flag for match channel 3.	0
4	CR0 Interrupt	Interrupt flag for capture channel 0 event.	0
5	CR1 Interrupt	Interrupt flag for capture channel 1 event.	0
6	CR2 Interrupt	Interrupt flag for capture channel 2 event.	0
7	CR3 Interrupt	Interrupt flag for capture channel 3 event.	0

Setting up & configuring Timers

- Set appropriate value in TxCTCR (Count Control Register ➔Set as Timer).
- Define the Prescale value in TxPR (Prescale Register) Set Value(s) in Match Register(s) if required.
- Set appropriate value in TxMCR if using Match registers / Interrupts.
- Reset Timer –Which resets PR and TC.
- Set TxTCR to 0×01 to Enable the Timer when required (Timer Control Register).
- Reset TxTCR to 0×00 to Disable the Timer when required (Timer Control Register).

Initialize Timer

```
T0CTCR = 0x00;T0PR = PRESCALE-1;
T0TCR = 0x02; //Reset Timer
```

Prescale (TxPR) Related Calculations:

PR+1 = Fin / Fout

For 1ms Tick

PR+1 = 60MHz/1KHz
= 60,000
PR = 59999

timer.H

- void initPLL(void);
- void delayms(int milliseconds);
- void delayus(int microseconds);